# Ch 6.2: Shrinkage - Ridge regression

## Lecture 17 - CMSE 381

Prof. Mengsen Zhang

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Oct 8, 2025

# Announcements

**Last time:**

- Subset selection

**This time:**

- Ridge regression

**Announcements:**

- HW #4 due Sunday 10/12

CMSE381_F2025_Schedule : Schedule

| | | | | | |
|---|---|---|---|---|---|
| | M | 9/22 | ***Project Day & Review*** | | |
| | W | 9/24 | **Midterm #1** | | |
| 12 | F | 9/26 | Leave one out CV | 5.1.1, 5.1.2 | |
| 13 | M | 9/29 | k-fold CV | 5.1.3 | |
| 14 | W | 10/1 | More k-fold CV | 5.1.4-5 | |
| 15 | F | 10/3 | k-fold CV for classification | 5.1.5 | |
| 16 | M | 10/6 | Subset selection | 6.1 | |
| 17 | W | 10/8 | Shrinkage: Ridge | 6.2.1 | |
| 18 | F | 10/10 | Shrinkage: Lasso | 6.2.2 | HW #4 Due Sun 10/12 |
| 19 | M | 10/13 | PCA | 6.3 | |
| 20 | W | 10/15 | PCR | 6.3 | |
| | F | 10/17 | ***Review*** | | |
| | M | 10/20 | Fall Break | | |
| | W | 10/22 | **Midterm #2** | | |
| 21 | F | 10/24 | Polynomial & Step Functions | 7.1-7.2 | HW #5 Due Sun 10/28 |
| 22 | M | 10/27 | Step Functions; Basis functions; Start Splines | 7.2-7.4 | |
| 23 | W | 10/29 | Regression Splines | 7.4 | |

Section 1

# Last time

# Subset selection

**Algorithm 6.1** *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly $k$ predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest $R^2$.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, $C_p$ (AIC), BIC, or adjusted $R^2$.

**Algorithm 6.2** *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest $R^2$.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, $C_p$ (AIC), BIC, or adjusted $R^2$.

**Algorithm 6.3** *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all $p$ predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all $k$ models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these $k$ models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest $R^2$.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, $C_p$ (AIC), BIC, or adjusted $R^2$.

# What should you learn from this lecture?

- What is regularization? Why do we need it?
- What are the two basic types of regularization methods? How are they implemented mathematically in linear regression?
- How do you fit a ridge regression model in python?
- How do you control the model flexibility & bias-variance tradeoff when using regularization?
- How do you find the right amount of regularization using cross-validation? How do you do this in python?
- What additional precautions do you need to take when using regularization (compared to least squares)?
- What are the advantages of regularization compared to Least Squares?
- What are the advantages of regularization compared to subset selection?

Section 2

# Ridge Regression

# Goal

- Fit model using all $p$ predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso

# Ridge regression

**Before:**

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

- $\lambda \geq 0$ is a tuning paramter to figure out separately.
- Second term is called "shrinkage penalty"
- $\lambda = 0$ means least squares estimate
- first encouter of regularization, $l_2$ norm, $l_p$ norm
- regularization = constrain = less flexibility

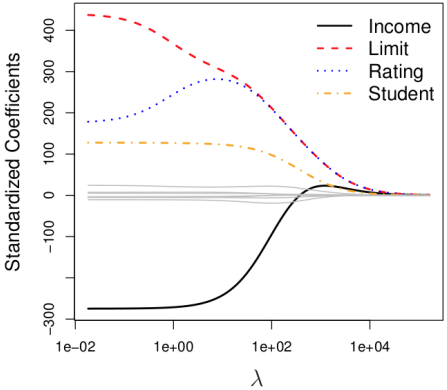
**After:**

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

- As $\lambda \to \infty$ impact of shrinkage penalty grows, ridge regression coeff will go towards 0
- Not applied to the intercept
- call coefficients found by ridge regression $\hat{\beta}_\lambda^R = (\beta_{1,\lambda}^R, \cdots, \beta_{p,\lambda}^R \cdots)$

# Example from the `Credit` data

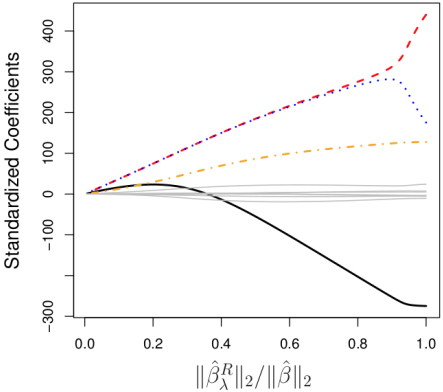
$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$




- Left x-axis, $\lambda \approx 0$
- Each line is the predicted coefficient value for a single variable
- as $\lambda$ increases everything goes to 0, basically giving the null model at the right end

# Same Setting, Different Plot

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2 \qquad \|\beta\|_2 = \sqrt{\sum_{j=1}^{p} \beta_j^2}$$




- $\hat{\beta}$ is the vector of least squares coeff estimates
- $\|\cdot\|$ measures distance from 0
- $\|\hat{\beta}_\lambda^R\|/\|\hat{\beta}\|$ ranges from 1 (when $\lambda = 0$) down to 0
- Amount the coefficient estimates have been shrunk
- more comparable across different regularization methods
- left more flexible, right less flexible

Test your understanding:PollEv

# Scale equivavariance (or lack thereof)

**Scale equivariant:** Multiplying a variable by $c$ ($cX_i$) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

- Ex: `income variable`.
- Least squares is scale equivariant
- Ridge regression very much is not
- $X_j\hat{\beta}^R_{j,\lambda}$ depends not only on $\lambda$ but also on values of other predictors

# Solution: Standardize predictors

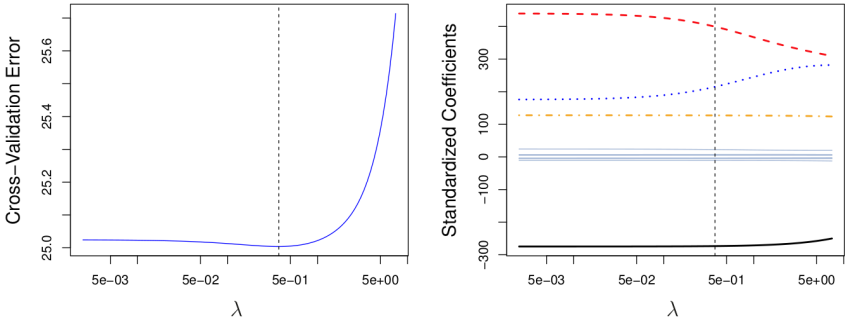
$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

- Denominator is estimated standard deviation of the jth predictor
- Standardized predictors will all have a standard deviation of one
- Previous figures showe standardized ridge regression coeffs on the $y$-axis

# Using Cross-Validation to find $\lambda$

- Choose a grid of $\lambda$ values
- Compute the ($k$-fold) cross-validation error for each value of $\lambda$
- Select the tuning parameter value $\lambda$ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

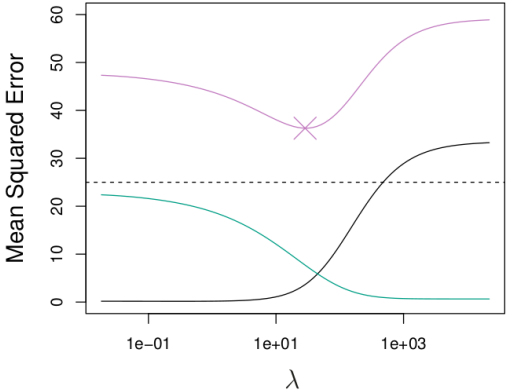
# LOOCV choice of $\lambda$ for ridge regression and `Credit` data




- Dashed vertical lines indicate the selected value of $\lambda$.
- In this case, small $\lambda$ so optimal fit involves small amount of shrinkage relative to least squares
- The dip is not very pronounced, so there is rather a wide range of values that would give a very similar error.
- In a case like this we might simply use the least squares solution.

# Coding

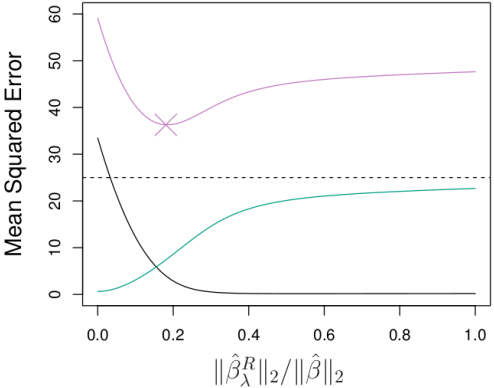
# Bias-Variance tradeoff




Squared bias (black), variance (green), and test mean squared error (purple) for simulated data. *Horizontal dashed line is minimum possible test MSE*

- Bias-variance tradeoff
- Minimum MSE achieved around $\lambda = 30$
- High variance means MSE for $\lambda = 0$ (least squares) and $\lambda = \infty$ (null model) are basically the same

# More Bias-Variance Tradeoff




Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

# Advantages of Ridge

**Ridge vs. Least Squares:**

- Previous slide and ability to lower variance
- Ridge regression works best in situations where the least squares estimates have high variance
- Can trade off small increase in bias for a large decrease in variance

**Ridge vs. Subset Selection:**

- So much better computationally!
- Only fits a single model rather than $2^p$
- Tricks available to solve all for all values of $\lambda$ simultaneous

# Look back and look ahead

CMSE381_F2025_Schedule : Schedule

| | | | | | |
|---|---|---|---|---|---|
| | M | 9/22 | ***Project Day & Review*** | | |
| | W | 9/24 | **Midterm #1** | | |
| 12 | F | 9/26 | Leave one out CV | 5.1.1, 5.1.2 | |
| 13 | M | 9/29 | k-fold CV | 5.1.3 | |
| 14 | W | 10/1 | More k-fold CV | 5.1.4-5 | |
| 15 | F | 10/3 | k-fold CV for classification | 5.1.5 | |
| 16 | M | 10/6 | Subset selection | 6.1 | |
| 17 | W | 10/8 | Shrinkage: Ridge | 6.2.1 | |
| 18 | F | 10/10 | Shrinkage: Lasso | 6.2.2 | HW #4 Due Sun 10/12 |
| 19 | M | 10/13 | PCA | 6.3 | |
| 20 | W | 10/15 | PCR | 6.3 | |
| | F | 10/17 | ***Review*** | | |
| | M | 10/20 | Fall Break | | |
| | W | 10/22 | **Midterm #2** | | |
| 21 | F | 10/24 | Polynomial & Step Functions | 7.1-7.2 | HW #5 Due Sun 10/28 |
| 22 | M | 10/27 | Step Functions; Basis functions; Start Splines | 7.2-7.4 | |
| 23 | W | 10/29 | Regression Splines | 7.4 | |

- What is regularization? Why do we need it?
- What are the two basic types of regularization methods? How are they implemented mathematically in linear regression?
- How do you fit a ridge regression model in python?
- How do you control the model flexibility & bias-variance tradeoff when using regularization?
- How do you find the right amount of regularization using cross-validation? How do you do this in python?
- What additional precautions do you need to take when using regularization (compared to least squares)?
- What are the advantages of regularization compared to Least Squares?
- What are the advantages of regularization compared to subset selection?